

LOCHANA MARASINGHE

Software Engineer and Doctoral Candidate at WSU, USA

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EDUCATION

Ph.D. in Biological and Agricultural Engineering

Washington State University, Pullman, WA, USA

Jan 2024 – Present

CGPA: 4.00/4.00

Postgraduate Diploma in Computer Science

University of Moratuwa, Sri Lanka

Jan 2021 – Aug 2022

OGPA: 3.75/4.20

B.Sc.(Hons) in Electrical and Information Engineering

University of Ruhuna, Sri Lanka

Jan 2016 – Mar 2020

OGPA: 3.64/4.00

Diploma in Software Engineering

IDM Achievers International Campus, Sri Lanka

Feb 2015 – Dec 2015

RESEARCH EXPERIENCE

Graduate Research Assistant

Department of Biological Systems Engineering, Washington State University

Jan 2024 – Present

Pullman, WA

Advisor: Sindhuja Sankaran (Ph.D.)

Manuscripts in Preparation:

Lochana Marasinghe, Arron H. Carter, Sindhuja Sankaran "IoT-Powered Phenomics: Development of Dynamic Crop Performance Monitoring System for Wheat Breeding Programs"

Technology Stack: Python, R, Java, Android Studio, Microsoft Azure, Google Play Console, Grafana, InfluxDB

Programmable Devices: Raspberry Pi, Arduino, ESP32, Data Loggers

Research Projects:

- Development and application of an IoT-based sensing system for continuous, high-throughput phenotyping in wheat breeding programs. [wheat-lens.azurewebsites.net]
- Evaluation of the IoT-based sensing system for its applications in a grower field for real-time monitoring and as a decision support tool. [Ongoing]
- Development of a LiDAR-based sensing system for automated early detection of potato diseases in commercial storage facilities.[Ongoing]
- Responsible for ongoing maintenance and feature development of an Android application designed to measure the length-to-width ratio of potato tubers using image processing and machine learning.[Ongoing]
- Development of an AI (Artificial Intelligence)-based system for accurate chickpea pod counting utilizing an automated ground robot. [Upcoming]

SKILLS

- Concepts:** OOP, SOLID, Agile, Data Structures and Algorithms, Design Patterns, ACID, RESTful API
- Core Technical Skills:** Java/Java EE, JavaScript, Typescript, SQL, HTML, CSS, PHP, Python, C++
- Frameworks & Libraries:** Spring Boot, Spring ORM, Spring Security, Angular, JADE, Node.js, Bootstrap, jQuery, OpenCV, scikit-learn, NumPy, Pandas, Android SDK, Android Studio
- Databases:** MySQL, Redis, Neo4j, Elasticsearch, InfluxDB
- Testing:** JUnit, Mockito, Postman
- Version Controlling:** Git, GitHub, GitLab
- Cloud Native:** Microsoft Azure, Google Cloud Platform, Amazon Web Services, Docker, Google Play Store
- Analytics:** JMeter, Grafana
- Hardware:** Arduino, NodeMCU, Raspberry Pi

INTERESTS

Robotics

Embedded Systems

IoT

Machine Learning

Computer Vision

Microservices

Competitive Programming

PROFESSIONAL

EXPERIENCE

Senior Software Engineer

CodeGen International

Jul 2022 – Jul 2023

Sri Lanka

Technology Stack: Java EE, Angular, TypeScript, Spring Boot, GCP

- Designed and implemented features in 5 products across 3 domains: education, web conferencing, and logistics.
- Completed designing, developing, and launching of vehicle delivery driver assigning system as a solution to minimize travel costs.
- Designed and developed authentication and authorization flows.
- Engaged in client interactions and created solutions to meet client requirements.
- Performed deployments of web applications on GCP App Engine and Compute Engine.
- Tested and monitored the performance of systems using JMeter and Grafana.

Software Engineer

CodeGen International

📅 Jul 2020 – Jun 2022

📍 Sri Lanka

Technology Stack: Java EE, Angular, JavaScript, Spring Boot, MySQL

- Collaborated on developing features in 2 products, including a learning management system for universities, and a service for conducting exams.
- Engineered an efficient appointment scheduling solution for individuals and organizations.
- Engaged in agile development processes, including design, planning, and estimation tasks.

Intern Software Engineer

Effective Solutions

📅 Jan 2019 – Apr 2019

📍 Sri Lanka

Technology Stack: Node.js, HTML, JavaScript, MySQL, Bit Bucket

- Performed database migration tasks using MySQL
- Contributed to the development of a sales management system for Ceylon Chocolates Ltd using Node.js.

Intern Software Engineer

Institute of Java and Software Engineering

📅 Nov 2017 – Mar 2018

📍 Sri Lanka

Technology Stack: Java, Hibernate, HTML, CSS, JavaScript, MySQL

- Developed an online game using HTML, CSS, and JavaScript.
- Collaborated on developing a point-of-sales system, which involved creating both a standalone application and a web application using Java, Hibernate, and PHP.

ACHIEVEMENTS



AgAID Digital AgATHON, Washington State University, USA. 2025

First Runners-up



XbotiX Inter-University Robotic Competition, Sri Lanka. 2017

First Runners-up



Sky of Icarus Inter-University Drone Competition, Sri Lanka. 2019

Second Runners-up



HackADev National Youth Software Competition, Sri Lanka. 2018

Finalist



HackX Inter-University Hackathon, Sri Lanka. 2018

Finalist



International Autonomous Robotics Competition (IARC) of Techkriti – IIT, Kanpur, India. 2018

Participant

PRESENTATIONS

Poster presentations

- Lochana Marasinghe, Arron H. Carter, Michael O. Pumphrey, Vishal Jeyam, Sindhuja Sankaran "IoT Powered Phenomics: Continuous Crop Performance Monitoring in Wheat Breeding" WSU Plant Science Symposium 2025.

PEER-REVIEWED

PUBLICATIONS

- Andriamihajaso, T.F., **Marasinghe, L.T.**, McGee, R.J., and Sankaran, S. 2026. Development and evaluation of a smart crop stress monitoring and management system for applications in controlled environment. *Discover Agriculture*, August 2025. [Minor Revision]
- **Marasinghe, L.T.**, Paiboonvorachai, C., Carter, A.H., and Sankaran, S. 2026. IoT-powered phenomics: Continuous crop performance monitoring for wheat breeding. *Smart Agricultural Technology*. [Internal Review]

COURSES

- **Fundamentals of Deep Learning** (NVIDIA)
- **AWS Academy Machine Learning Foundations Course** (AWS Academy)
- **Data Integration Specialist Course** (AWS Educate)
- **Cloud Computing 101 Course** (AWS Educate)
- **Intro to Machine Learning Course** (Kaggle)
- **Intermediate Machine Learning Course** (Kaggle)
- **Introduction and Programming with IoT Boards – Pohang University of Science and Technology** (Coursera)

MENTORING EXPERIENCE

- Mentor for summer undergraduate research interns in 2024, Phenomics Lab, Washington State University, WA, USA.
- Mentor for summer undergraduate research interns in 2025, Phenomics Lab, Washington State University, WA, USA.

VOLUNTEER EXPERIENCE

- **IEEE Eminence v1.0 2019, University of Ruhuna, Sri Lanka** – Inter-Faculty Women in Engineering Design Competition Organizing Committee
- **Shield of Code v1.0 2019, University of Ruhuna, Sri Lanka** – Inter-Faculty Hackathon Organizing Committee
- **SOS Charity Event (Mathematics Seminar Series) 2018, Sri Lanka** – Volunteer in teaching
- **IEEE Day 2017, University of Ruhuna, Sri Lanka** – Organizing Committee

EXTRA CURRICULAR

- **Secretary – IEEE Student Branch, University of Ruhuna, Sri Lanka** [June 2019 – June 2020]
- **Secretary – IEEE Industrial Application Society, University of Ruhuna, Sri Lanka** [June 2018 – May 2019]

OUTREACH

- Phenomics lab representative for the STEM research opportunity fair organized by the Office of Undergraduate Research, Washington State University, WA, USA. [October 2025]
- Phenomics lab/Department of Biological Systems Engineering representative for the Lab Crawl event organized by the College of Agricultural, Human, and Natural Resource Sciences, Washington State University, WA, USA. [February 2025]
- Phenomics lab representative for the STEM research opportunity fair organized by the Office of Undergraduate Research, Washington State University, WA, USA. [October 2024]

PROJECTS

Sleep Classification Using XeThru Sensor

Partnering with Novelda AS in Norway, this undergraduate project aimed to improve sleep disorder detection by accurately classifying respiration and movement signals in different sleep stages.

- Technology Stack:
 - Languages: Python, C++
 - Programmable Devices: NodeMCU ESP32, Raspberry Pi

EO/IR Based Maritime Surveillance System in Collaboration with Telecommunications Regulatory Commission of Sri Lanka

This project supports maritime surveillance in detecting violations of maritime policies such as maritime security, fisheries control, and smuggling by effectively identifying objects on the ocean surface.

- Technology Stack: Python, CNN – Single Shot Detector, AWS Sagemaker

StockBot: Stock Market Analysis Solution

An automated solution for the stock market investors to take better investment decisions through the analysis of technical indicators such as MACD, RSI, and Bollinger Bands. Investors can subscribe to companies of their choice, and receive technical indicator updates directly in their Twitter inbox via a Twitter chatbot. This project utilises SaaS applications of AWS.

- Technology Stack: AWS Lambda, AWS API Gateway, Auth0, Alpha Vantage API

Fingerprint Attendance Marking System

A fingerprint-based attendance tracking system for universities, allowing easy and accurate tracking with real-time access for both lecturers and students.

- Technology Stack:
 - Languages & Frameworks: PHP, HTML, JS, Bootstrap, AJAX
 - Programmable Devices: Arduino (C/C++)

Automatic Drug Injector

An automatic drug injection device with a syringe for patients requiring continuous low-dose medication treatment over a certain period of time.

- Technology Stack: Arduino (C/C++)

Drowsiness Detector

A drowsiness detection system for drivers to prevent accidents. Drowsiness is detected by calculating the eye-aspect ratio, then alerts the driver when the eye-aspect ratio is low.

- Technology Stack: Python, OpenCV, NumPy

Distributed Content Searching Network

A distributed file-searching network utilizing an unstructured peer-to-peer Gnutella network. Upon file request initiation, the file is searched via query flooding until the file is found. The search result is then sent back to the requesting peer, and if multiple peers have the file, it selects one to receive it.

- Technology Stack:
 - Languages: Java
 - Deployment: Amazon EC2 instances

SPORTS

- **Weightlifting Inter Faculty Novices' Competition, University of Ruhuna, Sri Lanka – First Runners-up [2018]**
- **Taekwondo Inter Faculty Sports Meet, University of Ruhuna, Sri Lanka – First Runners-up [2017, 2018]**